INCH-POUND

MIL-B-52239D
29 May 1992
SUPERSEDING
MIL-B-52239C
12 February 1985

MILITARY SPECIFICATION

BAR, CONNECTING, BRIDGE, STEEL:

FOR PONTON FLOAT, HALF SECTIONAL,

PNEUMATIC, 12-TON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a steel connecting bar to be used with a 12-ton, pneumatic, ponton float.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-R-52243	- Retainer, Bridge Pin.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 519	- Seamless Carbon and Alloy Steel Mechanical Tubing.
A 575	- Steel Bars, Carbon, Merchant Quality, M-Grades.
B 633	- Electrodeposited Coatings of Zinc or Iron and Steel.
D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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AMERICAN WELDING SOCIETY (AWS)

A2.4	- Symbols for Welding, Brazing and Nondestructive Examination.
A5.1	- Covered Carbon Steel Arc Welding Electrodes.
D1.1	- Structural Welding Code - Steel.
D1.2	- Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. AGENT

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Rm 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The connecting bar shall be as shown on figure 1 and as specified herein. The connecting bar shall be fabricated from one continuous length of tubing, with a steel flange at one end and a hemispherically shaped nosepiece at the other end, both of which shall be welded to the connecting bar.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. The material shall be as specified herein. Materials not specified shall be selected by the contractor, and shall be subjected to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The connecting bar shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the connecting bar may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.3.1 Tubing. The connecting bar shall be round seamless tubing conforming to ASTM A 519, grade 1015 to 1025, hot finished, annealed.

3.3.3.2 Bar steel. The connecting bar flanges and hemispherical nosepiece shall be steel conforming to ASTM A 575, grade M1015 to M1025.

3.3.3.3 Welding electrodes. Welding electrodes shall conform to AWS A5.1, E 7014 to E 7028.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the connecting bar may be newly fabricated from recovered materials to the maximum extent practicable, provided the connecting bar produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the connecting bar.

3.4 Retainer. The connecting bar shall be provided with a retainer conforming to MIL-R-52243, length 5-9/16 inches.

3.5 Treatment. The completed connecting bar shall be cleaned, treated, and zinc coated in accordance with ASTM B 633, type II, Fe/Zn 25 SC4.

3.6 Metal stamping. Each connecting bar shall be stamped as shown in figure 1, and as specified (see 6.2). Metal stamping shall be performed on the connecting bars prior to application of zinc coating and shall be legible after the coating has been applied.

3.7 Workmanship.

3.7.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners.

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3.7.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the connecting bar to conform to the examination requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch. Parts joined by groove welds shall be ground smooth and finished to the diameter of the parent metal. Unless otherwise specified (see 6.2), the welding process used in fabrication of the connecting bars shall be at the option of the contractor. Welding and welds shall be in accordance with either ASME or AWS code.

3.7.3 Welders and welding operators. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D1.1, D1.2 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.2 Examination. Sample selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The connecting bar shall be examined as specified herein for the following defects:

101. Dimensions not as specified (see 3.1).
102. Materials not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable operating and storage environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2.1).
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
106. Used, rebuilt or remanufactured components, pieces, and parts incorporated in the connecting bar (see 3.3.4).
107. Components missing or not as specified (see 3.4).
108. Zinc coating missing or not as specified (see 3.5).
109. Identification marking missing, incomplete, or not illegible (see 3.6).
110. Workmanship not as specified (see 3.7).
111. Welding and welds not as specified (see 3.7.2).
112. Welders and welding operators not certified as specified (see 3.7.3).
113. Welding certification not available as specified (see 3.7.3).

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

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4.6.1.1 Unit of product. For the purposes of inspection, a completely processed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. One or more defects shall be cause for rejection.

114.	Boxes not as specified (see 5.1.1 and 5.1.2).
115.	Strapping not located as specified (see 5.1.1 and 5.1.2).
116.	Strapping not zinc coated for level A (see 5.1.1 and 5.1.2).
117.	Marking illegible, incorrect, or incomplete (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The connecting bars with pins shall be packed in quantities of 12 in close-fitting boxes conforming to PPP-B-621, class 2, style 2 or PPP-B-601, overseas type, style optional. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable, and ASTM D 4675.

5.1.2 Level B. The connecting bars with pins shall be packed in quantities of 12 in close-fitting boxes conforming to PPP-B-621, class 1, style 2 or PPP-B-601, domestic type, style optional, and strapping may be zinc-coated.

5.1.3 Level C. The connecting bars with pins shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.2 Markings (see 6.4). In addition to any special marking as specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The connecting bar is intended to be used to fasten together two pneumatic ponton floats and to connect the outrigger beam to the ponton float.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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c. When a first article is required for inspection and approval, and the number of units required (see 3.2, 4.3 and 6.3).
d. Additional data required for metal stamping (see 3.6).
e. When the welding process will not be at the option of the contractor (see 3.7.2).
f. Level of packing required (see 5.1).
g. Any special marking (see 5.2).

6.3 First article. When a first article inspection is required, the connecting bar should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Levels of preservation. Levels of preservation have not been included, as the components do not require preservation. For the purpose of marking, the containers shall be marked A/A, A/B or A/C.

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractural application of the data requirements, a Contract Data Requirements List (DD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
a. 3.	DI-MISC-80875	Welding Procedures	
b. 3.	DI-MISC-80876	Welding Procedure Qualification Test Report.	

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.6 Subject term (key word) listing.

Bridge connecting bar
Connecting bar
Fastening bar
Ponton float connector
Steel connecting bar

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

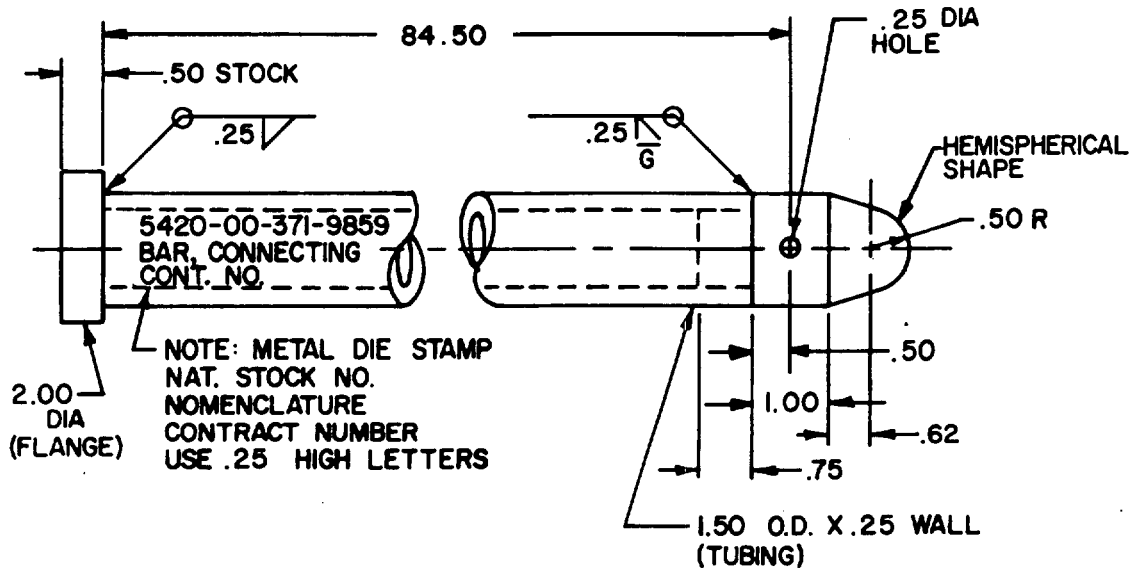
Custodian:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 5420-0190

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES
2. UNSPECIFIED TOLERANCE: ±.03
3. WELDING SYMBOLS ARE IN ACCORDANCE WITH ANSI/AWS A 2.4

FIGURE 1. Bar, connecting, bridge.

X2000B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-52239D	2. DOCUMENT DATE (YYMMDD) 92/05/29

3. DOCUMENT TITLE
Bar, Connecting, Bridge, Steel: For Ponton Float, Half Sectional, Pneumatic, 12-Ton

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD FORM 1426 OCT 89 Previous editions are obsolete

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